ELECTRONEGATIVITY 9

The number of electrons in the outer shell (valency) of an atom of an element affects the number and type of bonds this element will form with other elements. The electronegativity of an atom within a compound also affects the types of bonds formed. Electronegativity describes the attraction of an atom for electrons.

Electronegativity is the power of an atom to attract electrons.

1 Use a textbook to give another definition of *electronegativity*.

Non-metals with a strong 'desire' to gain electrons and complete their outer electron shells have high values of electronegativity. Metals with a strong 'desire' to lose electrons to form full outer shells have low values of electronegativity.

2 Explain how a metal can lose electrons to form a full outer shell (octet).

Generally, small atoms, especially those with nearly-full outer shells of electrons, attract electrons more easily than larger atoms do. Hence small atoms tend to have higher electronegativity values. The following generalisations can be made about bonding between atoms:

- When there is little or no difference in electronegativities a covalent bond is formed
- When there is a moderate difference in electronegativities a polar covalent bond is formed
- When there is a considerable difference in electronegativities an ionic bond is formed.

Figure 1 represents how electrons are 'shared' in bonds. The greater the size of the shaded area, the greater the share an atom has of a bonding electron. In polar covalent bonding the δ (delta) sign means **relatively**. Hence one end of the molecule is relatively positive (δ+) to the other end (δ−).



Figure 1

Table 1 shows the symbol (**a**), atomic number (**b**) and electronegativity (**c**) of some elements.

a	H	Li	Be	B	C	N	O	F	Na	Mg	Al	Si	P	S	Cl	K	Ca
b	1	3	4	5	6	7	8	9	11	12	13	14	15	16	17	19	20
c	2.1	1.0	1.5	2.0	2.5	3.0	3.5	4.0	0.9	1.2	1.5	1.8	2.1	2.5	3.0	0.8	1.0

Table 1

3 Refer to table 1 to answer the following.

- **a** The values for helium, neon and argon are not quoted in table 1. Suggest why.
- **b** Plot a graph of atomic number (x-axis) versus electronegativity (y-axis).
- **c** What happens to the electronegativity values as you go across a Period?
- **d** What happens to the electronegativity values as you go down a Group?

4 Use the electronegativities in table 1 to predict whether the compounds formed between the following elements would be covalent, polar covalent or ionic.

- **a** potassium and chlorine
- **b** carbon and sulfur
- **c** hydrogen and water

5 Mark each statement as true or false.

- **a** The most electronegative elements are the reactive non-metals in the top left-hand of the Periodic Table.
- **b** The symbol δ means 'a positive charge'.
- **c** Non-metals tend to form ionic compounds with metals.

© Peter Gribben, Wendy St Germain and Blake Education 2005
ISBN 978 1 92561 066 6